

ECE 4644 / 5610 Satellite Communications

Homework Assignment #8 SOLUTIONS

Problem #1

- 6 /
- a) In the baseband of a transmission link the symbols (e.g., rectangular pulses, impulses) are sharply delineated in time, with one symbol ending before another begins. With passage through an RC low-pass filter, the symbol information becomes spread in time and the symbols lose their distinct identity. In several sentences and with the help of a rough sketch, explain how this comes about and relate it to the occurrence of *inter-symbol interference* (ISI).
 - b) Filtering is seen to be the cause of ISI. Why is it not practical to avoid ISI altogether by running the communication link without filtering?
 - c) Explain how setting the entire link transfer function equal to a Nyquist raised cosine (RC) filter copes with the problem. Make a sketch and use the terms *waveform*, *tails*, *sampling times*, and *zero ISI*.

Problem #2

A baseband data link transmits NRZ input pulses and is designed to create a zero ISI waveform in the receiver with an overall raised cosine (RC) transfer function.

- 6 /
- a) For a symbol rate of 100 MHz, plot the one-sided filter characteristic of the raised cosine transfer function (TF) for roll-off rates of $\alpha = 0, 0.25$, and 0.5 . Plot using a computer program or draw carefully (and realistically) by hand.
 - b) Explain why communication links are designed with filters at both the Tx and Rx ends that have the same *root raised cosine* (RRC) transfer function. Hint: several reasons can be cited.
 - c) For a satellite link with matched RRC filters and the three combinations of symbol rate and roll-off rate given in part (a), find the occupied bandwidth of the link and the noise bandwidth. (Remember that this is a baseband data link.)

Problem #3

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A Ku-band satellite downlink has carrier frequency of 14.125 GHz and carries a symbol stream at $R_s = 16$ Msps. The transmitter and receiver have ideal RRC filters with $\alpha = 0.25$. Determine the bandwidth occupied by the RF signal (i.e., the *channel bandwidth*) and the frequency range of the transmitted RF signal. (Observe that this is an RF data link.)

Problem #4

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A satellite transponder has a bandwidth of 36 MHz. A group of earth stations has ideal SRRC filters with $\alpha = 0.5$. Determine the maximum bit rate that can be sent through this transponder with each of these types of modulation:

- a) BPSK
- b) QPSK
- c) 8-PSK

Problem #5

Answer each of these questions in 2-3 well-written sentences and with sketching where appropriate.

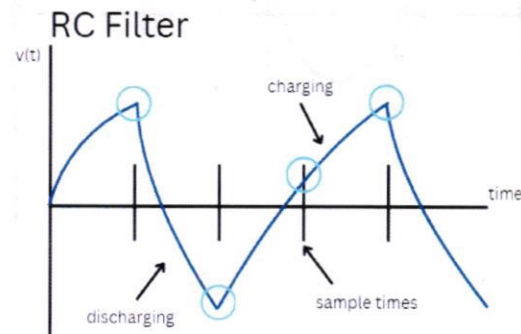
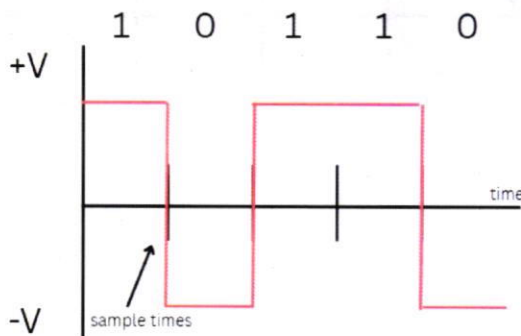
- a) Explain why and how equalization is typically performed on the Tx end of a link to deal with NRZ pulses as input.
- b) Ideally the transfer function associated with the propagation medium, $L(f)$, would be *flat* with respect to amplitude and *linear* with respect to phase. In your own words explain the meaning of these two terms. If this transfer function is not flat, how do we compensate in the design of the link?
- c) With reference to a BPSK demodulator, describe how the modulating bits are recovered by mixing, filtering, and sampling the output of the IF section (no math required but sketching would be appropriate).

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$\zeta = 26$

Problem #1.

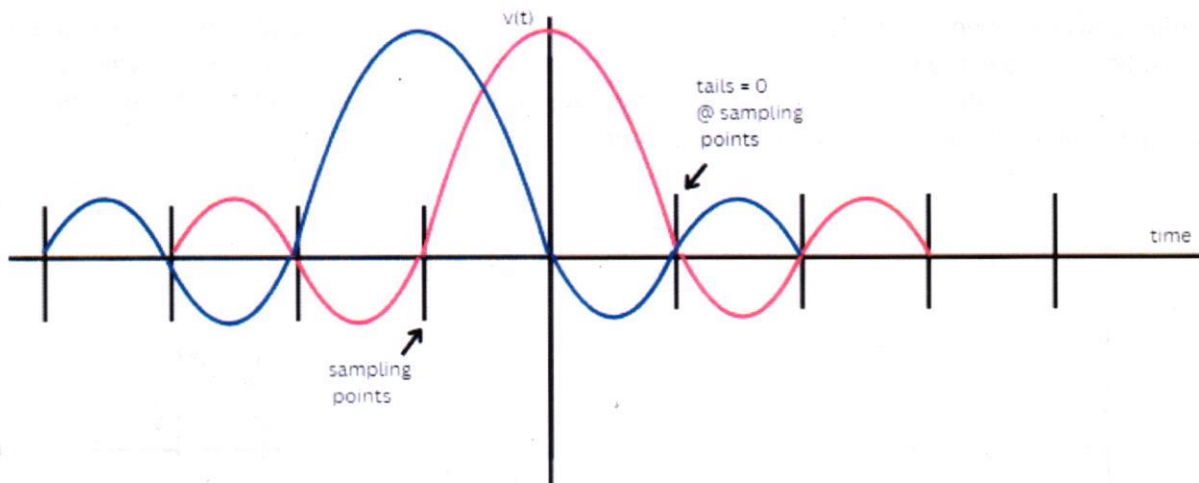
a) From electronics, we learned that capacitors do not charge and discharge instantaneously – this charging delay (RC time constant effect) seen in RC low-pass filters results in the baseband's individual symbols bleeding into one another. Essentially, the time that spaced out the symbols before (or the period for each pulse) is not enough to space out the capacitor's need to discharge and charge without interfering into another pulse period. You no longer have enough time for a clear delineation of symbols and instead they meld into one another. When a pulse trails into the next pulse period, the melding explained above, this is called Inter-Symbol Interference (ISI). If you were to take a sample the result would not just be the pulse from the period you're looking at but also the trailing charge or discharge of its neighbor as well. As noted in the figure below, the signals are delayed and lose their clean delineation of voltage values – making it more likely you'll make an error reading the pulse value. This could result in samples being wrong, even more so in the presence of noise.



b) It is not feasible to run the communication link without filtering because this would require infinite bandwidth to retain the pulse shape. While removing filtering might avoid ISI, it is not possible for wavelengths to be both time and frequency limited. Links have to operate within a finite bandwidth, smaller bandwidths are more efficient and allow for more communication systems to operate as there is a finite amount of bandwidth space. Filtering allows us to limit the bandwidth and is necessary.

c) Using a Nyquist raised cosine filter would result in a Rx pulse having a sinc shape. With this, we are forcing the waveform to be zero at every sample period except for when the waveform has maximum amplitude. While tails still extend past their period, the waveform at different sampling times is zero so you only get the value of the pulse you're interested in without input from its neighbor, essentially the tails are at zero. These qualities constitute a Zero ISI waveform because the tails and therefore wave bleeding into periods does not have an effect on sampling. From the figure below you can see that the value of the red waveform at $t=0$ is not affected by the value of the blue waveform. Similarly, at the peak of the blue waveform, the value of red does not disrupt the sampling because its tail is zero.

Zero ISI Waveform



Problem #3.

We can calculate the occupied bandwidth with the formula for an RF data link $B_{occ} = R_s(1 + \alpha)$.

$B_{occ} = 16(1 + .25) = 20 \text{ MHz}$. We know that bandwidth is centered in the frequency range so we will need to calculate the min and max frequencies. We can calculate the minimum and maximum frequencies with the equations $f_{min} = 14.125 \text{ GHz} - 10 \text{ MHz} = 14.115 \text{ GHz}$ and $f_{max} = 14.125 \text{ GHz} + 10 \text{ MHz} = 14.135 \text{ GHz}$. Therefore the frequency range is 14.115 to 14.135 GHz. ✓

Problem #4.

$B_{occ} = 36 \text{ MHz}$, $\alpha = .5$, $R_b = ?$, $R_s = ?$

First, we need to find the symbol rate. We can use the following equation: $R_s = \frac{B_{occ}}{(1+\alpha)}$.

$$R_s = \frac{36 \text{ MHz}}{(1+.5)} = 24 \text{ Msps.}$$

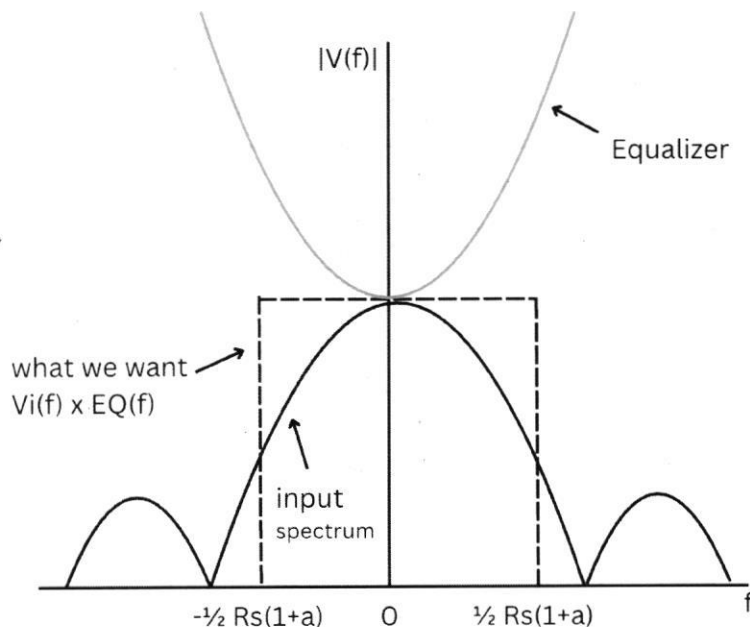
a) Since this is BPSK and it has one bit per symbol $R_b = 24 \text{ Mbps}$.

b) QPSK has two bits per symbol so $R_b = 48 \text{ Mbps}$.

c) 8-PSK has 3 bits per symbol so $R_b = 72 \text{ Mbps}$. ✓

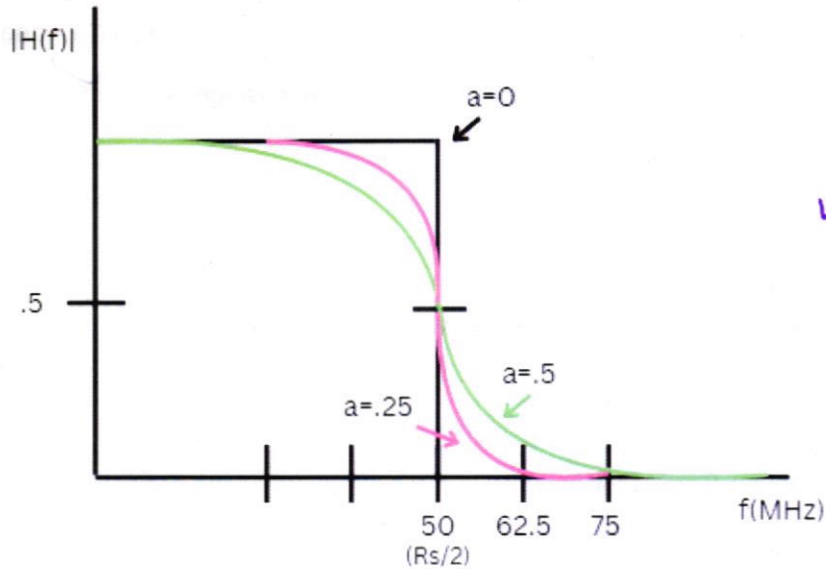
Problem #5.

a) Equalization is typically performed on the Tx end of a link to for NRZ pulses because they are square and to satisfy the Nyquist RC transfer characteristic an impulse is necessary. Essentially, the spectrum of NRZ pulses is $\sin X/X$ and we need the input spectrum to be flat. To achieve this we add an input equalizer with the frequency response $x/\sin x$ to create the flat spectrum. ✓



Problem #2.

a)



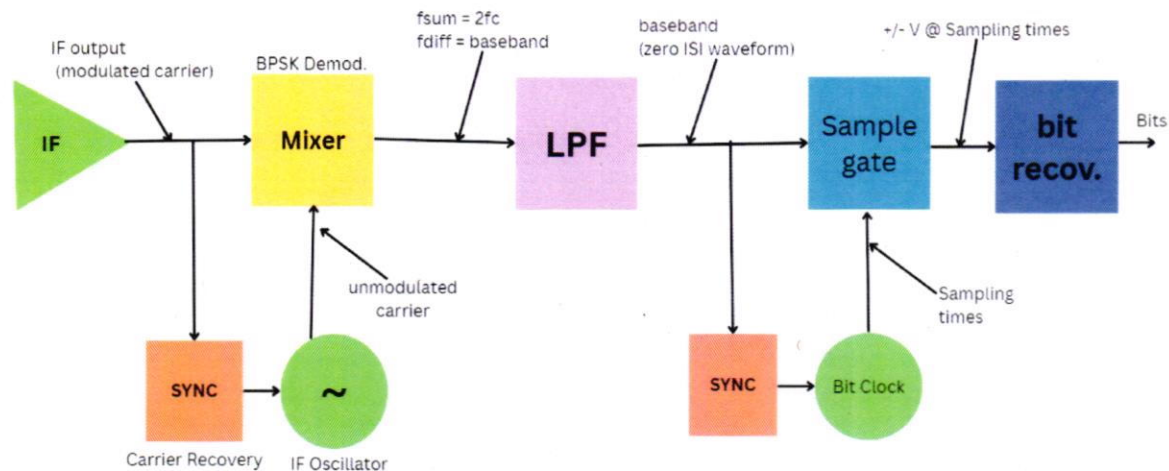
b) We need filtering at both ends, on the transmit end for bandwidth control and on the receiving end for noise control – it makes sense to put it at each end so that it still combines into a raised cosine transfer characteristic. Plus, both the Tx and Rx ends having the same RRC is considered a Matched Filter (MF). A link that uses a MF produces the best C/N ratio possible. Additionally, an RRC qualifies as a Zero ISI waveform design and would produce a zero ISI time waveform.

c)

α	Occupied bandwidth (B_{occ})	Noise Bandwidth (B_N)
0	$B_{occ} = \frac{1}{2}R_s(1 + \alpha) = \frac{1}{2}100(1 + 0) = 50\text{MHz}$	$B_N = \frac{1}{2}R_s = 50\text{MHz}$
.25	$B_{occ} = \frac{1}{2}R_s(1 + \alpha) = \frac{1}{2}100(1 + .25) = 62.5\text{MHz}$	$B_N = \frac{1}{2}R_s = 50\text{MHz}$
.50	$B_{occ} = \frac{1}{2}R_s(1 + \alpha) = \frac{1}{2}100(1 + .5) = 75\text{MHz}$	$B_N = \frac{1}{2}R_s = 50\text{MHz}$

b) The ideal requirements on the transfer function $L(f)$ are a flat amplitude response of $|L(f)| = 1$ and a linear phase response of $\phi(f) = kf$. The flat amplitude would result from the medium the wave is traveling through not distorting the signal at all. Essentially, all the frequencies we're sending's amplitude go down at the same rate and there's no unevenness, the receiving frequencies will be lower but they will all be in the same proportions of the transmitted frequencies. The phase being a linear factor of the frequency essentially means that none of the frequencies should be altered or distorted. For example a doppler shift only applying to some frequencies. If these conditions are not met, an equalizer can be inserted to compensate. $EL(f)$, i.e., $L(f) \times EL(f) = 1$.

c)



First, we need to strip out the carrier wave. The first sync block recovers the IF carrier from the noisy BPSK signal and locks in the IF oscillator, here an unmodulated version of the carrier is created by combining the carrier with itself. Then the mixer is used to mix the modulated and unmodulated waves to produce the sum and difference where the difference is the baseband frequency and the sum is at frequency of $2f_c$. The lowpass filter is then used to filter out the sum of the mixer output and leave only the baseband frequency. Following the low pass filter you have a DC output, a zero ISI waveform. The second sync block locks to the bit rate of signal and synchronizes the bit clock that drives the sample gate. This ensures that samples of the wave are taken at the original sampling times. With the baseband waveform and the sampling times, the sample gate takes a screenshot of the waveform at the sample time and sends the value of the voltage captured to the next block. The symbol recovery block outputs the digital symbols by converting the voltages from the sample gate to 0s or 1s from negative or positive voltages respectively.